Informed Trading on Capitol Hill: A Quantitative Portfolio Construction Approach to Congressional Stock Disclosures

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1 Introduction

> "We're a free market economy, they [members] should be able to participate in that"
> *—Former U.S. Speaker of the House Nancy Pelosi, December 15, 2021 (CNBC 2021)*

Less than a year after this statement, Pelosi disclosed millions of dollars of purchased NVIDIA stock, three weeks before the passing of the CHIPS Act and with it a subsequent surge in NVIDIA stock price. The CHIPS Act, legislation Pelosi was a part of, aimed over $240 billion dollars to semiconductor and other tech research in the United States. Whether this trade was just a coincidence or not, it highlights the current tension around congressional stock trading: the same individuals writing and voting on legislation that has the power to impact markets are permitted, and in Pelosi's opinion, entitled to trade on the stocks those markets contain.

The academic research on this topic is divided. Eggers and Hainmueller (Eggers and Hainmueller 2013) find that the average congressional portfolio underperforms a passive index by 2-3% annually, concluding that there is no evidence of widespread systematic insider trading. However, Karadas (Karadas 2019), which analyzes the same era, finds that members serving on the most influential committees earned abnormal annual returns exceeding 35% under short-term holding periods. These gains, which were primarily concentrated among inexperienced traders, strongly imply access to time-sensitive private information rather than investment skill. Importantly, both studies find these abnormal returns vanished following the passage of the STOCK Act in 2012, legislation that forced public disclosure of congressional trades. Considering both previous studies, these findings suggest that the congressional trading signal is small but relevant. Beneath a large population of uninformed or average traders is a smaller subset of members whose trades are exceptionally well-timed and financially successful.

Our prior research (Barker and Ichiba 2025) reinforces this view using post-STOCK Act data from 2012 to 2023. While average congressional returns were statistically insignificant compared to the S&P 500 benchmark, a persistent group of outlier members generated gains far beyond those of even the top professional fund managers, consistent with the pattern Karadas identifies among those powerful committee members. That finding warranted an appropriate follow-up: can these top performing trades be identified systematically before the fact, and can a portfolio constructed from this signal generate persistent out-of-sample excess returns?

This study attempts to answer both questions using methods drawn from quantitative finance. We construct a weighted alpha signal from CAPM-adjusted cumulative abnormal returns (CARs) across 15,278 congressional purchase events from 2012 to 2026, applying exponential time decay and multi-member corroboration filters to isolate the most informative trades. We then clean the resulting covariance matrix using Random Matrix Theory eigenvalue decomposition under the Marchenko-Pastur law, and construct a mean-variance optimal portfolio using convex optimization. A walk-forward backtest from 2015 to 2025 evaluates out-of-sample performance against SPY.

We found an answer to the first question, and a partial answer to the second. While blindly copying every congressional stock purchase won't generate statistically significant abnormal returns (CAR-30 = 0.054%, p = 0.489), not every trade is equal as some members are consistently more successful. Purchases made by members in the top tercile of historical trading performance generate CAR-30 of +1.39% (t=9.18, $p < 0.001$), and coordinated purchases (multiple members buying the same stock within a 30-day window) generate CAR-30 of +0.24% (t = 2.20, p = 0.028). The resulting portfolio follows the SPY closely over the backtest period (mean annual excess returns = -0.93%), with meaningful outperformance in four of nine years and severe underperformance in 2022 attributable more to the selected stocks in large growth sectors - where information advantages are stronger - which declined sharply after the Federal Reserve's rate hikes, rather than poor portfolio selection.

2 Data

The congressional trading data used in this study was an aggregation of the trading data used in our previous study, combined with additional trades collected from the Senate EFTS electronic filing system, as well as new data from the Lambda Finance congressional trading API. The additional trades were able to fill in major gaps with the House disclosures as the House disclosures website presents many challenges when trying to obtain the data directly. Even with more data present in this study, our availabilty of House data still remains more sparse.

The daily price data was obtained from the yfinance API, and the FRED database provided the risk-free rate (3-month T-bill daily rates). The SPY ETF was used as the market benchmark as it is directly tradable and has daily price data available in yfinance.

Table 1: Sample construction and filtering pipeline

Step	N	Dropped	Reason
Data assembly			
Raw combined records	137,157	–	5 sources merged
After deduplication	61,270	-75,887	Overlap across sources
Transaction filtering			
Purchases only	30,204	-31,066	Sales, exchanges removed
Equity instruments only	22,297	-7,907	Bonds, funds removed
Valid ticker	22,083	-214	Unparseable ticker
Price and beta estimation			
Buy-day price available	15,505	-6,578	No price on event date
Min. estimation window	15,278	-227	<60 days for beta
Final sample	**15,278**	–	

Table 2: CAPM-adjusted CAR summary statistics (n = 15,278)

Horizon	Mean	Std Dev	Median	Min	Max
CAR-10	−0.028%	5.74%	0.00%	−115.97%	76.52%
CAR-20	+0.019%	7.73%	0.00%	−61.28%	109.47%
CAR-30	+0.054%	9.68%	0.00%	−110.13%	191.79%
CAR-60	+0.179%*	12.95%	0.31%	−174.76%	217.31%

* p <0.10. 25th and 75th percentiles omitted for space.

In Table 1, the combined data sources were de-duplicated, filtered for only purchases and only those of valid stocks before matching prices to the buy days. This left a sample of $15,278$. Table 2 presents summary statistics for the CAPM-adjusted CARs across all four horizons.

It is important to understand, before proceeding, that key wording the in STOCK Act names a 45 day disclosure window for congress members. This can pose a challenge for researchers as prices can drastically fluctuate in that window leading to largely different transaction and disclosure prices. Additionally, some members may disclose their trades months late, only incurring a $200 fine for doing so. Our signal uses transaction dates for more accurate CAR computation, and this lag to disclosure time can be a limitation. Additionally, one should note that trades are disclosed in predetermined ranges (on the disclosure forms), not exact amounts. Thus we map these weights using a geometric midpoint and a log transformation. The geometric mean is calculated from the range of endpoints, and the log of that is taken to compress the scale.

3 Methodology

3.1 CAPM-Adjusted Cumulative Abnormal Returns

In our study, each congressional stock purchase is treated as an "event" at time equal to zero. A 252 (trading days in a calendar year) day estimation window is constructed, ending 30 days before the event to estimate β using ordinary least squares. On the daily, abnormal returns (AR) are calculated:

$$AR_t = R_{stock,t} - [R_{f,t} + \beta_{stock}(R_{market,t} - R_{f,t})]$$

Where $R_{f,t}$ is the risk-free rate at time t, and $R_{market,t}$ is the daily market return. The cumulative abnormal returns (CAR) were then calculated by summing the daily ARs over respective 10,20,30,60 day windows. A minimum of 60 observations was required for β estimation.

3.2 Excess Return Signal Construction

Using our CARs, we can calculate our expected returns, μ, using a weighted average of CAR-30 per ticker across all events. The three weighting components are: member quality (historical mean CAR), cluster signal (unique members buy the same ticker within 30 days), and trade size (mapped from the amount ranges). Additionally, we apply exponential time decay to ensure that trades from years ago will matter less than trades from last week. We also applied minimum filters, where each ticker is only included if it has three or more members buying it, and there are three or more events. Rather than a hard cutoff, extreme signal values are compressed using tanh winsorization, which smoothly dampens outliers while preserving the relative ranking of all tickers.

3.3 Covariance Matrix Estimation

We construct the empirical covariance matrix from daily log returns spanning 3,230 trading days. Ledoit-Wolf shrinkage is applied first to regularize the estimate and ensure matrix invertibility, particularly important given the large number of tickers relative to observations in earlier backtest windows.

We then apply Random Matrix Theory (RMT) denoising via the Marchenko-Pastur distribution (Marčenko and Pastur 1967) to separate genuine correlation structure from noise. For a random matrix with N assets and T observations, the ratio $q = N/T$ determines the theoretical bounds $[\lambda^-, \lambda^+]$ on the eigenvalue distribution that is expected under pure noise:

$$\lambda^\pm = \sigma^2 \left(1 \pm \sqrt{q}\right)^2$$

Eigenvalues falling within these bounds are statistically indistinguishable from noise and are replaced by their bulk mean, while eigenvalues exceeding λ^+ are preserved as they represent genuine covariance structure. The result is a cleaned, positive definite covariance matrix Σ.

3.4 Mean-Variance Portfolio Optimization

Portfolio weights are determined by solving the following mean-variance optimization problem:

$$\max_w \ w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$$

subject to $\sum_i w_i = 1$, $w_i \geq 0$ (long positions only), and $w_i \leq 0.05$ (maximum 5% per position). The risk aversion parameter $\lambda = 1.0$ is held fixed throughout. The optimization is solved using the `cvxpy` package (Diamond and Boyd 2016).

3.5 Walk-Forward Backtest

To evaluate out-of-sample performance we employ a walk-forward backtest from 2015 to 2025. For each year Y, both μ and Σ are estimated using only data available prior to January 1st of year Y, ensuring no look-ahead bias. The optimized portfolio is then held for one year and its return compared against a passive investment in the S&P 500 ETF that serves as the standard market return benchmark. The backtest begins in 2015 rather than 2012 to allow the 2012-2014 years of STOCK Act disclosures to accumulate as sufficient estimation data.

4 Results

4.1 Hypothesis Test Results

We test the null hypothesis that mean CAPM-adjusted cumulative abnormal returns equal zero across all congressional purchase events. Table 3 presents results across four holding horizons for the full sample of all 15,276 purchases.

Table 3: CAPM-adjusted CAR t-tests

Horizon	**Mean CAR**	**Std Dev**	**t-stat**	**p-value**	**Sig.**
CAR-10	−0.029%	5.74%	−0.618	0.536	
CAR-20	+0.018%	7.73%	+0.292	0.770	
CAR-30	+0.054%	9.68%	+0.683	0.494	
CAR-60	+0.179%	12.95%	+1.713	0.087	*

* p <0.10, ** p <0.05, *** p <0.01

Consistent with Eggers and Hainmueller (2013), undifferentiated congressional purchases generate no statistically significant abnormal returns at standard confidence levels. Table 3 and Figure 1 show the only exception is CAR-60, which is marginally significant at $\alpha = 0.10$ ($t = 1.713$, $p = 0.087$), suggesting that any informational advantage takes time to be reflected in prices, consistent with the up-to-45-day disclosure lag required under the STOCK Act.

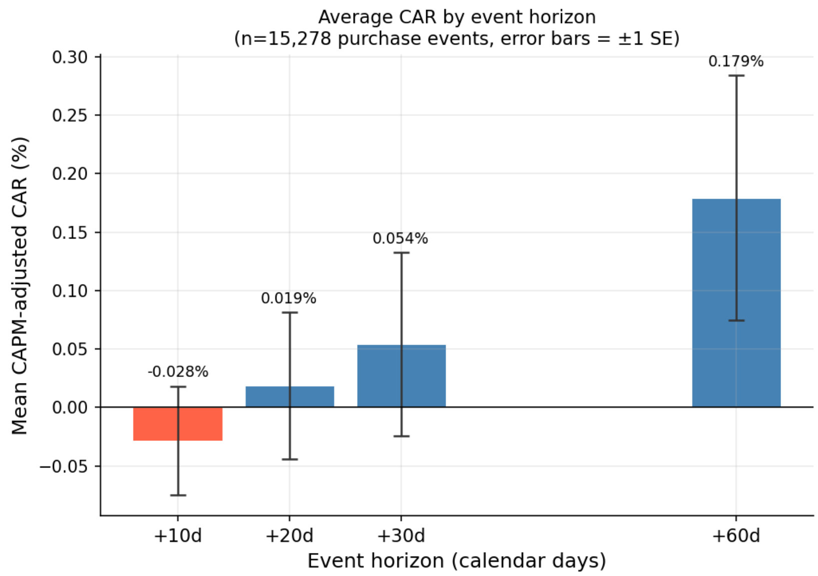

Figure 1: CAR Significance with Standard Errors

Table 4: CAR-30 subgroup t-tests (H_0: mean CAR-30 = 0)

Subgroup	**n**	**Mean CAR-30**	**t-stat**	**p-value**	**Sig.**
Member quality (historical mean CAR-30)					
Top tercile	5,267	+1.390%	+9.164	<0.001	***
Bottom tercile	5,118	−1.325%	−10.206	<0.001	***
Coordinated buying (same ticker, within 30 days)					
Cluster trades	7,414	+0.241%	+2.201	0.028	**
Non-cluster trades	7,862	−0.123%	−1.106	0.269	
Ticker conviction (unique members buying same ticker)					
High conviction (≥4 members)	12,039	+0.086%	+1.095	0.274	
Min-threshold (3 members)	1,026	+0.291%	+0.762	0.446	

* p <0.10, ** p <0.05, *** p <0.01

Table 4 shows members in the top tercile of historical trading performance generate mean CAR-30 of +1.390% ($t = 9.164$, $p < 0.001$), while bottom tercile of members generate −1.325% ($t = -10.206$, $p < 0.001$), a spread of 2.72 percentage points over 30 days. This result strongly validates the member quality weighting component of our methodology, confirming that informational advantages are concentrated among a smaller subset of members rather than distributed across the legislature.

Coordinated purchases which are defined as multiple members independently buying the same stock within a 30-day window, and these also generate significantly positive abnormal returns of +0.241% ($t = 2.201$, $p = 0.028$), while isolated purchases do not ($p = 0.269$). This suggests that when multiple legislators purchase the same stock, possibly reacting to the same information, the trade carries a stronger signal of genuine informational advantage. Notably, the number of unique members ever buying the same stock is not significant, indicating that the timing of coordinated purchases matters more than the total count of buyers.

4.2 Portfolio Construction

After proceeding with the process described in our methodology, we were left with 633 tickers, of which 73 had more than 20% of their price data missing, and thus were dropped. This created a log-return matrix of 3,230 days by 560 tickers, with a Marchenko-Pastur $q = N/T = 0.1734$. Our bounds for the Marchenko-Pastur distribution are $[0.000158, 0.000930]$, with 30 real signal eigenvalues explaining 60.1% of the variance.

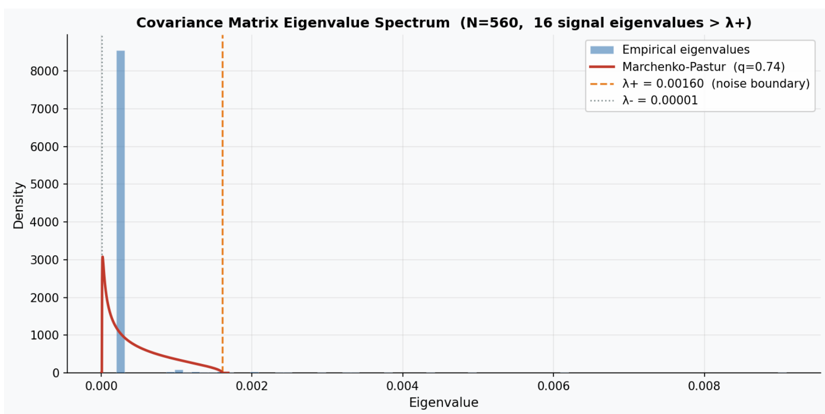

Figure 2: Covariance matrix eigenvalue spectrum

In Figure 2, the blue histogram shows the empirical distribution of eigenvalues from the Ledoit-Wolf shrinkage estimator. The red curve is the theoretical Marchenko-Pastur distribution under pure noise.

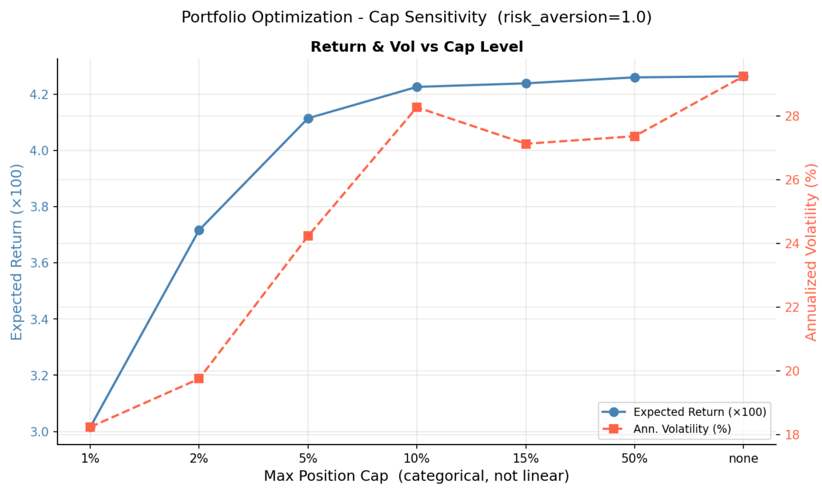

Figure 3: Efficient Frontier of Portfolio

Figure 3 shows the cap-constrained efficient frontier across position limits ranging from 1% to unconstrained. As the cap becomes more relaxed, expected return increases which reflects the optimizer's ability to concentrate more weight into the highest-μ stocks. Annualized volatility, however, behaves differently: it rises from about 18% at the 1% cap to a peak of about 29% in the unconstrained case, but declines sharply to $\tilde{2}7\%$ at the 15% and 50% cases. This suggests that the covariance matrix identifies a subset of high-signal stocks with sufficiently low pairwise correlations. That moderate concentration actually improves the risk-return tradeoff relative to the 10-position portfolio. For the primary analysis we use a 5% maximum position cap, giving a 20-position portfolio that balances signal concentration with diversification.

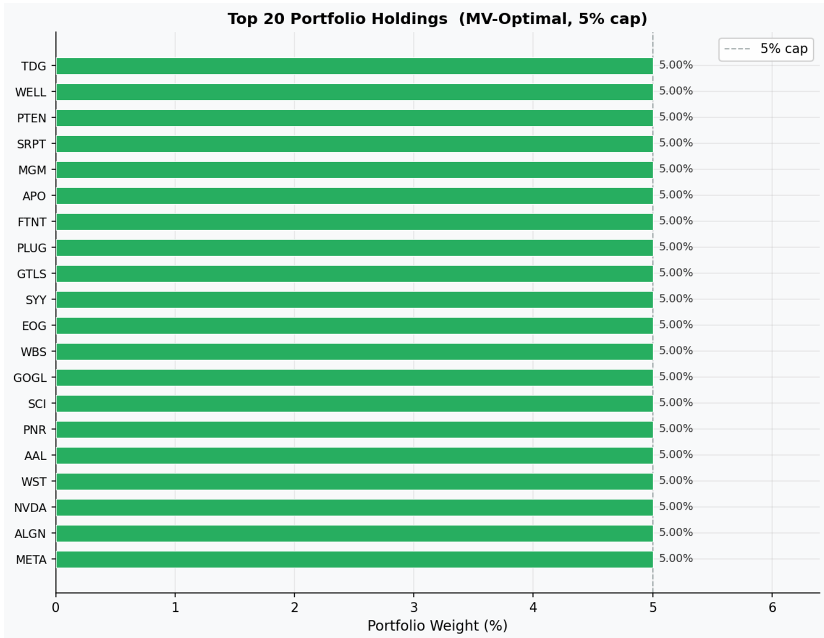

Figure 4: Top 20 Holdings by Portfolio Weight

Figure 4 shows the resulting holdings. The portfolio naturally clusters around sectors where congressional informational advantages could be the strongest: defense and aerospace (TDG, GTLS), energy (EOG, PTEN), biotech and healthcare (SRPT, WELL), cybersecurity (FTNT), and financial services (APO, WBS). This sector clustering emerges purely from the weighted CAR signal without any explicit sector constraints, providing support for the idea that legislative committee oversight creates informational advantages in specific industries.

4.3 Backtest Performance

Table 5: Walk-forward backtest results

Year	Portfolio Return	SPY Return	Alpha
2016	+14.79%	+13.59%	+1.20%
2017	+18.43%	+20.78%	−2.35%
2018	+0.53%	−5.25%	+5.78%
2019	+30.07%	+31.09%	−1.02%
2020	+19.23%	+17.24%	+2.00%
2021	+40.29%	+29.63%	+10.66%
2022	−36.83%	−19.95%	−16.88%
2023	+20.37%	+24.81%	−4.44%
2024	+17.40%	+20.70%	−3.30%
Mean	+13.81%	+14.74%	−0.93%

Metric	Strategy	SPY
Mean annual return	+13.81%	+14.74%
Mean alpha	−0.93%	–
Mean alpha ex-2022	+1.07%	–
Years outperforming SPY	4 / 9	–
Sharpe ratio (rf = 2%)	0.541	0.756
Max drawdown	−36.83%	−19.95%

Table 5 shows the year-by-year walk-forward backtest results from 2016 to 2024. The strategy generates a mean annual return of 13.81% against SPY's 14.74%, yielding a mean annual alpha of −0.93% over the full period. The strategy outperforms SPY in 4 of 9 years, with its strongest years being 2021 (+10.66% alpha) and 2018 (+5.78% alpha).

The most significant underperformance by far, occurs in 2022, where the strategy trails SPY by 16.88%. This is possibly due to the portfolio's concentration in growth-oriented sectors such as defense, biotech, and energy technology, all of which suffered disproportionately during the Federal Reserve's aggressive rate hiking cycle. As discussed in Section 3, the congressional trading signal tends to identify stocks in sectors subject to legislative oversight, which are typically growth-oriented. This creates an unhedged exposure that is unrelated to the quality of the underlying signal. Excluding 2022, the mean annual alpha improves to +1.07%, suggesting the strategy generates genuine excess returns over more normal market years.

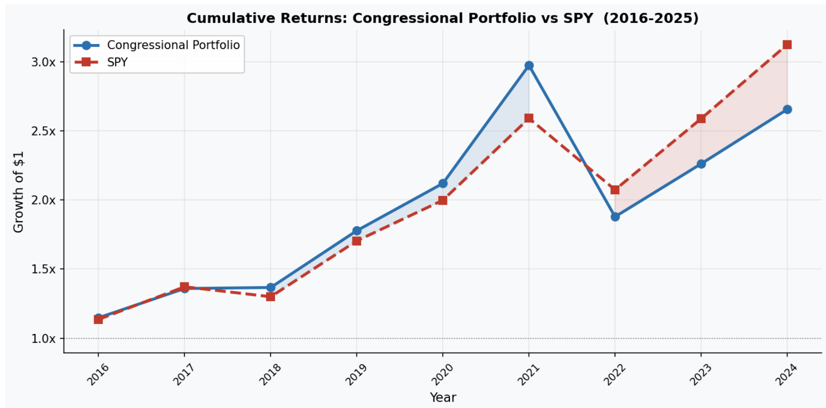

Figure 5: Cumulative Returns: Senate Portfolio vs SPY

Figure 5 shows the cumulative growth of $1 invested in the strategy versus SPY from 2016 to 2024. Observe the congress portfolio tracks SPY closely to 2020, outperforms substantially in 2020 and through most of 2021, then dipping below and underperforming through 2024.

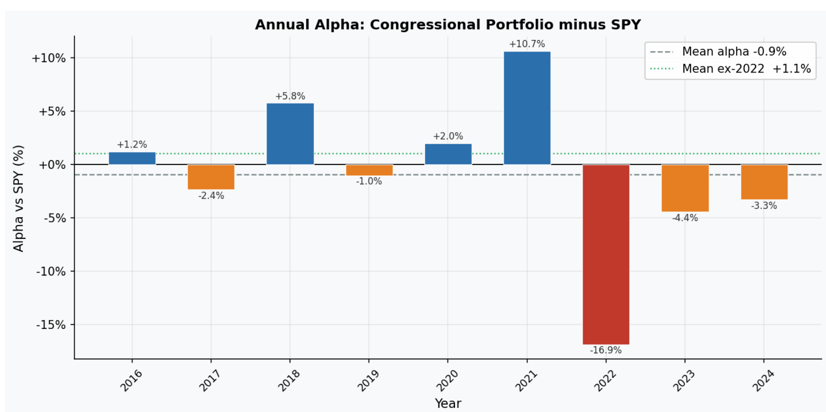

Figure 6: Annual Alphas from 2016 to 2024

Figure 6 shows the year-by-year alpha, with the 2022 drawdown clearly dominating the nine-year mean. The portfolio Sharpe ratio of 0.541 compared to SPY's 0.756 (Table 5) reflects this same pattern where the strategy takes on more volatility than the market without generating enough additional return over the full sample to compensate. Excluding 2022, however, the mean alpha of +$1.07% and outperformance in years such as 2018 (+5.78%) and 2021 (+10.66%) suggest the signal generates genuine excess returns in more normal market years.

5 Policy Implications

The findings of this study do not establish that any member of Congress has engaged in illegal trading activity. The STOCK Act prohibits members from trading on material and non-public information obtained through their legislative duties, and enforcement of that prohibition remains the responsibility of appropriate authorities. However, the statistically significant abnormal returns observed among top-performing members and in coordinated purchases are consistent with the hypothesis that some trades reflect informational advantages arising from legislative access. To the extent that this is the case, several policy measures could reduce the potential for such advantages to manifest in personal trading activity.

The most direct intervention would be a mandatory blind trust requirement, which would require members to transfer their investment portfolios to independently managed trusts upon taking office. This would prevent members from making individual stock selections while in office, eliminating the mechanism by which legislative information could translate into personal trading gains. The Transparent Representation Upholding Service and Trust (TRUST) Act was introduced by U.S. representatives Seth Magaziner (D) and Chip Roy (R) in 2025, a bill that would mandate blind trusts (Magaziner and Roy 2025). The act would require all members of congress to enter such a blind trust within 180 days of the bills passing, and require certification of doing so by the House and Senate, with the process of those certifications being public record. However, blind trusts could still suffer from a member advocating for legislation they know has been placed in the trust, loading up on certain stocks before entering office knowing the trust will hold them, or influencing the trustee indirectly through relationships or some form of coercion.

Shortening the disclosure window from 45 days to a shorter period, such as 5 to 10 days, would increase market transparency and reduce the window during which undisclosed trades could benefit from non-public information. Our results suggest that the congressional trading signal weakens at shorter horizons, with CAR-60 showing marginal significance, while CAR-10 does not, which is consistent with the idea that reducing the disclosure lag would compress the window of informational advantage. Karadas (2019) similarly finds that abnormal returns on congressional portfolios are highest under a one-week holding period and disappear entirely beyond three months, consistent with the time-sensitive nature of legislative information. This short-term concentration of returns also provides empirical support for reducing the disclosure window, as the informational advantage is largely exhausted before the current 45-day requirement elapses.

Finally, stronger enforcement of existing STOCK Act provisions, including higher penalties for late disclosure and greater resources for investigating suspicious trading patterns around legislative events, could prove most feasible in terms of implementation, and be highly effective. Current STOCK Act enforcement provisions impose a maximum fine of $200 for late disclosure, a figure that bears no meaningful relationship to the potential gains from delayed reporting documented in the literature. With existing enforcement being widely characterized as insufficient, bills like the Ending Trading and Holdings in Congressional Stocks (ETHICS) Act aims to strengthen the penalties, and expand on the blind trust idea. The act would again force congressional members to enter blind trusts, but seeing the flaws in such a system, would include provisions requiring divestiture of the assets entering the trust (Krishnamoorthi 2025). Those in violation of the act, including their family members, would be fined at a minimum their monthly pay. Beyond existing proposals, policymakers could further strengthen deterrence by applying insider trading penalties similar to SEC civil provisions, fines of up to $1,000,000 or three times the value of the trade (*15 U.S.C. § 78u-1: Civil Penalties for Insider Trading* 2024), to cases involving suspicious trading patterns around legislative events.

6 Conclusion

This study assessed whether high-performing congressional trades can be identified systematically before the fact, and whether a portfolio constructed from that signal generates persistent out-of-sample excess returns. The evidence suggests a partial yes to both questions. The signal exists, as evidenced by statistically significant abnormal returns among top-performing members and coordinated trades, but it does not produce a portfolio that consistently beats the market benchmark. Instead, excess returns appear concentrated among a small subset of congressional traders who, when acting in coordination on the same stock within a short window, generate high returns consistent with shared access to time-sensitive legislative information.

This was apparent in the overall CARs being insignificant across most horizons, while member quality and coordinated cluster trades were highly significant ($p < 0.001$ and $p = 0.028$ respectively). The constructed portfolio approximately tracked SPY over the backtest period, with a mean annual alpha of -0.93%, influenced heavily by the extreme downturn in 2022. Excluding that year, the mean annual alpha improves to $+1.07\%$, suggesting the signal generates genuine excess returns in normal market years.

In assessing these results, several limitations are worth noting. Most significant is that our purchase data is predominantly from Senate disclosures, as the structure of the House disclosure system made data collection substantially more difficult. We also assume transaction costs are negligible, which is unrealistic in practice but outside the scope of this study. The strategy's performance is sensitive to the macroeconomic environment, tending to outperform when growth stocks are rewarded and underperform when rising interest rates compress growth valuations. Also, signal construction parameters were selected using the full sample, which may introduce some look-ahead bias and exaggerate out-of-sample performance.

Future research could address these limitations in several ways. Additional House trading data would increase sample size and diversify the information captured, while collection is technically feasible using disclosed document numbers, it would require automated PDF parsing of individual filings. More complete data, combined with committee- specific weighting, could improve the portfolios performance. Most promisingly, applying this same pipeline to SEC Form 4 corporate insider disclosures represents a natural extension as corporate insiders face a two-day rather than 45-day disclosure window, trade more frequently, and span a broader universe of stocks. This could potentially yield a stronger and more consistently exploitable signal.

7 References

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